

Exam 1 Solution Manual 2 DE

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1 Problem 1

$$\frac{dy}{dx} = \frac{xy^3}{\sqrt{1+x^2}}$$

1.1 Solution

Lets use Separation of Variables to solve

$$\frac{1}{y^3} dy = \frac{x}{\sqrt{1+x^2}} dx$$

Integrate

$$\int \frac{1}{y^3} dy = \int \frac{x}{\sqrt{1+x^2}} dx$$

$$\frac{-1}{2y^2} = (1+x^2)^{1/2} + K_1$$

$$y^2 = \frac{-1}{2((x^2+1)^{1/2} + K_1)}$$

Let's keep y^2 as is.

2 Problem 2

$$y' = \frac{2y^4 + x^4}{xy^3}$$

2.1 Solution

Let's solve as a homogeneous DE

$$\frac{dv}{dx}x + v = \frac{2x^4v^4 + x^4}{x^4v^3}$$

Simplify and rearrange for integration

$$\frac{v^3}{v^4 + 1} dv = \frac{1}{x} dx$$

Integrate

$$\frac{1}{4} \ln(v^4 + 1) = \ln|x| + K_1$$

Do some algebra and get rid of v

$$\left(\frac{y^4 + x^4}{x^4}\right)^{1/4} = xK_2$$

Final answer

$$(y^4 + x^4)^{1/4} = x^2 K_2$$

3 Problem 3

$$y' = \frac{2 + ye^{xy}}{2y - xe^{xy}}$$

3.1 Solution

Let's solve this as an exact DE. First, test for exactness.

$$\frac{\partial M}{\partial y} = 0 - e^{xy} - xye^{xy}$$

$$\frac{\partial N}{\partial x} = 0 - e^{xy} - xye^{xy}$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

The DE is exact

$$\int M(x, y) dx = \int -2 - ye^{xy} dx$$

$$F = -2x - e^{xy} + h(y)$$

$$\frac{\partial F}{\partial y} = 0 - xe^{xy} + h'(y)$$

Now set $\partial F_y = N$

$$h'(y) = 2y$$

Now integrate

$$h(y) = y^2$$

Plug into F

$$F = -2x - e^{xy} + y^2 + K_1$$

4 Problem 4

$$xy' = 3y + x^2$$

4.1 Solution

Let's solve this as a linear DE.

$$\frac{dy}{dx} = \frac{3y}{x} + x$$

Find the integrating factor

$$IF = e^{\int \frac{-3}{x} dx} = \frac{1}{x^3}$$

$$\int d(x^{-3}y) = \int \frac{1}{x^2} dx$$

$$y = -x^2 + K_1 x^3$$

5 Problem 5

$$\sin(x)y' + \cos(x)y = 2x$$

5.1 Solution

$$\frac{dy}{dx} + \frac{\cos(x)}{\sin(x)}y = \frac{2x}{\sin(x)}$$

Find the integrating factor.

$$IF = e^{\int \frac{\cos(x)}{\sin(x)} dx} = \sin(x)$$

$$\int d(y \sin(x)) = \int (2x) dx$$

$$y = \frac{x^2}{\sin(x)} + \frac{K_1}{\sin(x)}$$

6 Problem 6

A 300-gallon tank initially contains 120 gallons of pure water. Starting at $t = 0$, salt water containing 5.0 pounds of slater per gallon falls into the the tank at 4 gal/min. The thoroughly mixed solution flows out at 2 gal/min. How much salt is in the tank when the tank overflows?

6.1 Solution

$$\frac{dx}{dt} = 20 - \frac{2x}{120 + 2t}$$

$$\frac{dx}{dt} + \frac{2x}{120 + 2t} = 20$$

Find the integrating factor

$$IF = e^{\int \frac{2}{120+2t} dt} = (120 + 2t)$$

$$\int d((120 + 2t)x) = \int 20(120 + 2t)dt$$

$$(120 + 2t)x = 240t + 20t^2 + K_1$$

Find K_1

$$K_1 = 0$$

Find t when the tank overflows

$$300 = 120 + 2t$$

$$t = 90$$

Plug into equation and solve

$$x = \frac{20(120(90) + (90)^2)}{300}$$

7 Problem 7

At $t = 0$, a 17-pound object is pushed beneath the surface of a calm lake and then released from rest. The object has a buoyancy force of 21 pounds, so it begins to rise after release. The resistance of the water (in pounds) is given by $\frac{v^2}{4}$, where v is the upward velocity of the object in feet per second. If the object reaches the surface in 7 seconds, what is the velocity of the object when it reaches the surface of the lake?

7.1 Solution

$$21 - 17 - \frac{v^2}{4} = \frac{17}{32} \frac{dv}{dt}$$

$$-(v^2 - 16) = \frac{17}{8} \frac{dv}{dt}$$

$$\int \frac{-8}{17} dt = \int \frac{1}{v^2 - 16} dv$$

Split the integral

$$\int \frac{1}{(v-4)(v+4)} dv = \int \frac{A}{v-4} + \frac{B}{v+4} dv$$

$$A = \frac{1}{8}, \quad B = \frac{-1}{8}$$

$$\int \frac{-8}{17} dt = \frac{-8t}{17}$$

$$K_2 e^{-8t/17} = \left(\frac{v-4}{v+4}\right)^{1/8}$$

Get rid of constant

$$K_2 = \frac{(-4)^{1/8}}{(4)^{1/8}}$$

Plug in $t = 7$ and solve for v

$$(-1)e^{\frac{(-64)7}{17}} = \frac{v-4}{v+4}$$

After doing the algebra...

$$v = \frac{-4e^{(-64)7/17} + 4}{1 + e^{(-64)7/17}}$$